

Chapter 07

Threat Actors

Script Kiddies

Unskilled attackers

Use existing tools and scripts

Do not fully understand tools

Motivated by ego or recognition

Hacktivists

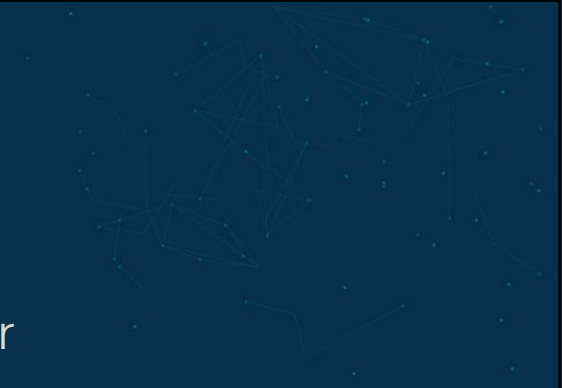
Motivated by political or social causes

Target organizations or institutions

Common attacks

- Defacement
- Data leaks
- DoS attacks

Driven by activism goals



Organized Crime

Financially motivated groups

Highly sophisticated attacks

Activities include

- Data theft
- Extortion
- Identity theft

Focus on profit

Advanced Persistent Threats (APT)

State-sponsored attackers

Highly advanced techniques

Long-term targeting

Motivated by national interest

Competitors as Threat Actors

Seek market advantage

Target rival organizations

May steal information

May disrupt services

Insider Threats

Originate within organization

May be malicious or accidental

Disgruntled employees possible

Can cause internal damage

Shadow IT Risks

Unauthorized IT usage

Not managed by IT team

Can impact security
and availability

Introduces unknown risk

Chapter 07

ttps

TTPs and Attribution

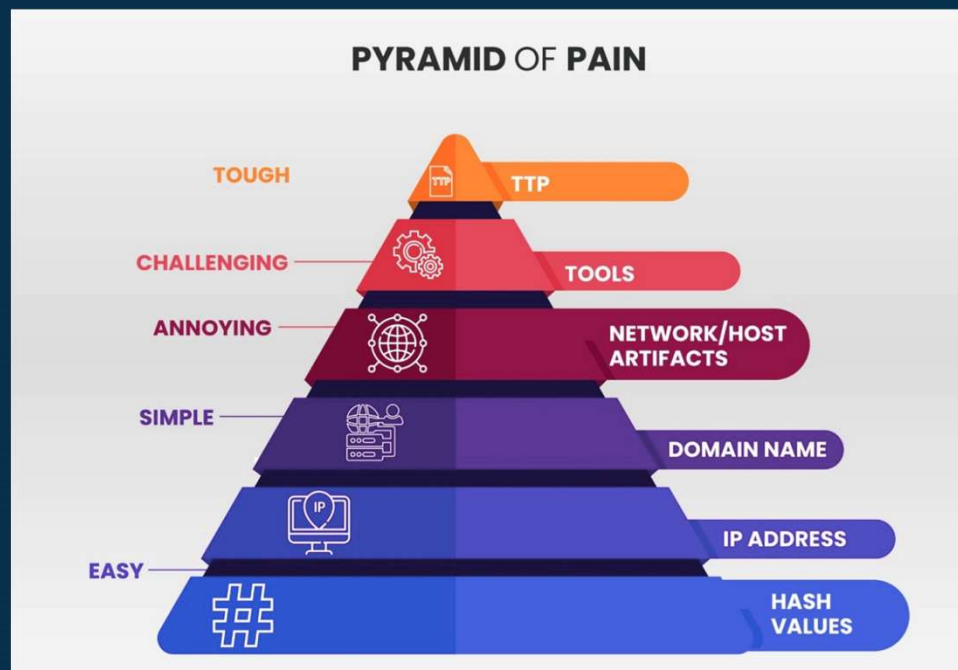
Link actions to threat actors

Identify attack patterns

Track adversary behavior

Support incident analysis

Pyramid of Pain



Pyramid of Pain

Hierarchy of attack indicators

Lower levels easier to change

Higher levels harder to evade

Focus on attacker disruption

Indicators vs TTPs

Lower indicators

- IPs hashes easily changed

Higher indicators

- Tools and TTPs harder to evade

Higher disruption to attackers

Focus on long-term impact

MITRE ATT&CK Framework

MITRE ATT&CK Tactics in the Enterprise Matrix



MITRE ATT&CK Framework

Knowledge base of
adversary behavior

Maps real-world attack patterns

Based on TTPs

Used for analysis and defense

Using ATT&CK

Identify potential adversaries

Map activity to patterns

Analyze systems and processes

Support defensive planning

Attribution Using TTPs

Link attacks to groups

Use behavioral clues

Identify campaigns

No claim needed

TTP Behavioral Patterns

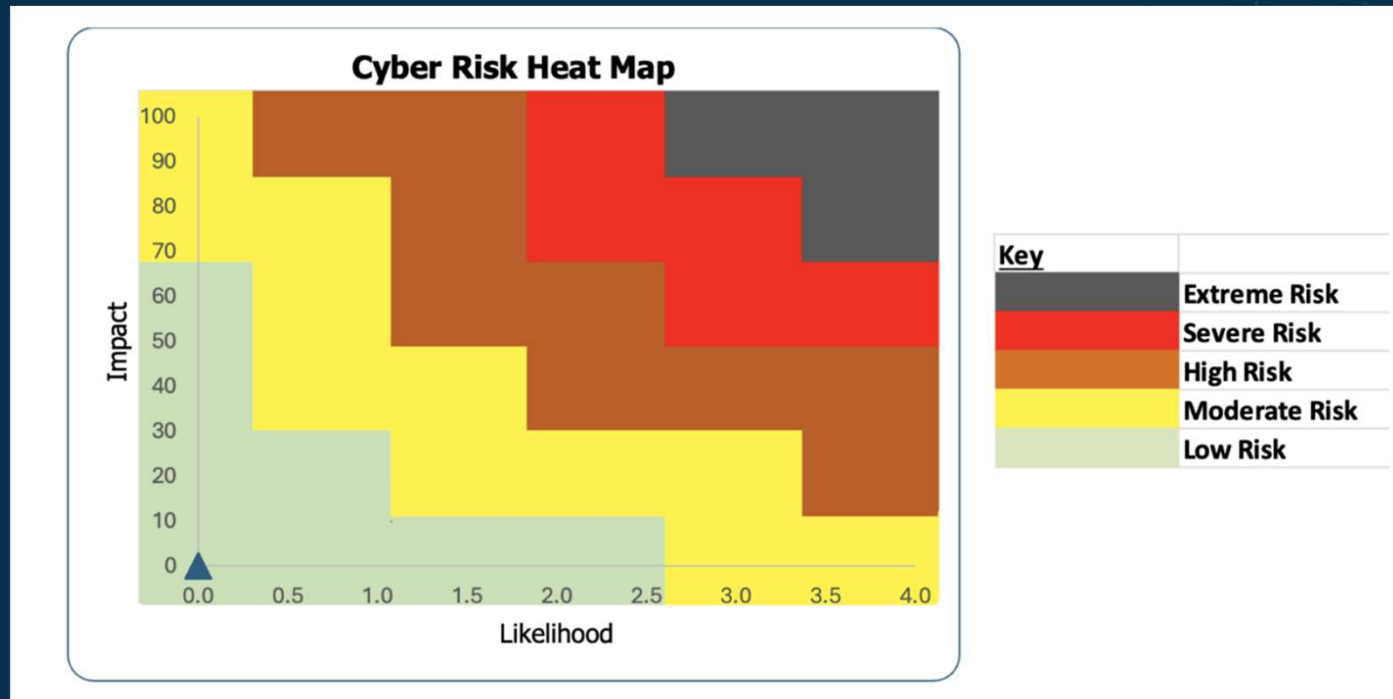
Unique attacker behaviors

Repeated identifiable actions

Create recognizable signatures

Used to track actors

Heat Map



Heat Map Analysis

Visualizes attacker behavior

Predicts likely next steps

Shows detection coverage

Identifies security gaps

TTPs in Defense Strategy

Use Intelligence from
past attacks

Identify likely adversaries

Build proactive defenses

Improve threat response

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Intel Sources

Open-Source Intelligence (OSINT)

Publicly available information

Accessible to anyone

Wide range of sources

Collected from open web

OSINT Challenges

Large volume of data

Duplicate information common

Requires validation effort

Must separate fact from rumor

Closed Source Intelligence

Subscription-based services

Paid intelligence sources

Pre-analyzed information

Delivered to members

Closed Source Benefits

Reduces false information

Provides actionable data

Less manual analysis required

Improves efficiency

Intelligence Sharing Organizations

Share intelligence across groups

Support collaboration

Improve situational awareness

Enhance security response

ISAC, ISAO, and SIX

ISAC

- Sharing and analysis center

ISAO

- Industry collaboration

SIX

- Information exchange groups

All support intelligence sharing

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Intel Metrics

Intelligence Evaluation Metrics

Intelligence must be evaluated

Uses defined metrics

Applied before analysis

Ensures usefulness of data

Relevance of Intelligence

Pertains to organization systems

Applies to current environment

Matches system versions

Determines operational impact

Accuracy of Intelligence

Measures reliability of data

Distinguishes rumor vs verified

Trusted sources
improve accuracy

Higher accuracy increases value

Timeliness of Intelligence

Determines if intelligence
is usable

Evaluates response window

Identifies outdated information

Supports timely action

Timeliness Example and Impact

Early warning before attack

Time to prepare defense

Improves response readiness

Enables preventive action

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Cyber Threat Intelligence

What Is Cyber Threat Intelligence?

Lessons learned from past attacks, applied to future defense

Turns suspicious activity into actionable insight

No CTI = flying blind

Data Enrichment

Adds context: geo data, TTPs, IPs, targeted systems

Raw data in, actionable intelligence out

Every enriched piece builds the bigger security picture

Intelligence Sharing Platforms

Platforms: OTX, MISP, OpenCTI

Feeds IOCs directly into SIEMs and vulnerability scanners

Today's threat to them is tomorrow's threat to you

Internal Reports and STRIDE

Org-specific reports from logs,
network data, and threat feeds

STRIDE: Spoofing, Tampering,
Repudiation, Info Disclosure, DoS,
Privilege Escalation

Categorize the threat → pick the
right control

Cyber Threat Maps

Live global attack visualizations
(FortiGuard, Kaspersky)

Show source, destination, and
attack type in real time

You never want to see yourself
on one

Deception Tools

Honeypots: decoys that flag any traffic as suspicious

Honeynets: entire fake networks of VMs as bait

Observe attacker TTPs, then share intel with the community